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DDARes–U^{2}Net: a dual-decoder adversarial residual U^{2}Net algorithm for segmentation of COVID-19 pneumonia lesions

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ABSTRACT.

Purpose: We aim to overcome the remaining bottlenecks in COVID-19 lesion segmentation from chest CT—namely, blurred lesion boundaries, false-positive responses from vessels or trachea, and the extreme variability of lesion shape, size, and location—by developing a dual-decoder adversarial residual U^{2}Net algorithm.

Approach: DDARes-U^{2}Net, an encoder–dual-decoder architecture, simultaneously performs both lung parenchyma segmentation and lesion segmentation within a unified network. The dual-decoder design incorporates a lung-parenchyma decoder that guides the model to focus specifically on the lung region, eliminating the information loss caused by traditional presegmentation approaches and markedly reducing computational overhead. Furthermore, an edge-loss function is introduced during training to enhance the network's ability to capture fine-grained lesion boundaries. Comparative experiments are conducted under two settings: within-dataset evaluation and cross-dataset evaluation. Ablation studies are performed to validate the contribution of each architectural component.

Results: The proposed DDARes-U^{2}Net model, with a relatively modest parameter count of 30.74 M, achieves superior generalization performance and segmentation accuracy compared with state-of-the-art methods. Across both within-center and cross-center experiments on five CT datasets (I to V), DDARes-U^{2}Net consistently ranks first for Dice and IoU. The largest gain is observed when the model trained on Dataset IV is tested on Dataset II, delivering $+3.41$ pp Dice versus ARes-U^{2}Net$++$ and $+5.51$ pp versus ARes-U^{2}Net, confirming its superior generalizability and clinical applicability.

Conclusions: The unified dual-decoder framework with explicit edge-aware supervision provides accurate, robust, and generalizable COVID-19 lesion segmentation, achieving the highest Dice and IoU in both intradataset and cross-center tests on five independent CT cohorts.

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Keywords: DDARes-U^{2}Net model; dual-decoder network; edge loss function; COVID-19 lesion segmentation

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1 Introduction

The coronavirus disease 2019 (COVID-19) outbreak that emerged in early 2020 has evolved into one of the most significant global public health emergencies in modern history, distinguished by

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its rapid transmission, extensive societal impact, and unprecedented challenges in containment.[1,2] As of March 2025, the pandemic has resulted in over 770 million confirmed cases and more than 7 million deaths worldwide, profoundly affecting human health, economic stability, and political landscapes across the globe. Although the World Health Organization has declared the end of COVID-19 as a Public Health Emergency of International Concern, research in this domain remains critically important for the imaging diagnosis of future emerging respiratory infectious diseases.

Automated lesion segmentation based on chest computed tomography (CT) images has emerged as a pivotal tool for assisted diagnosis and treatment evaluation, enabling efficient quantification of infected regions, precise localization of pathological changes, and reliable assessment of disease progression—thereby substantially enhancing diagnostic accuracy and therapeutic efficiency. However, accurate segmentation of COVID-19 lesions from CT images continues to face numerous challenges: image artifacts such as vascular and tracheal shadows interfere with precise detection, lesion boundaries are frequently indistinct, and intensity and texture features lack sufficient discriminability. Furthermore, the small size of some lesions leads to severe class imbalance at the pixel level, whereas conventional algorithms often suffer from detail loss during upsampling operations, limited receptive field design, high computational complexity, and insufficient segmentation accuracy.

With the advancement of deep learning techniques, U-Net[3] has established itself as a cornerstone model in medical image segmentation due to its encoder–decoder architecture and skip connections, which effectively integrate multilevel semantic and detailed information. A series of high-performance segmentation networks have been developed based on this foundation, including Attention U-Net[4] that incorporates attention gates to emphasize task-relevant features, residual U-Net[5] that integrates residual connections to enhance generalization capability, and U-Net++[6] that employs architecture search and dense connections to achieve multiscale feature fusion. Moreover, approaches that stack multiple U-Net modules have been proposed to improve detail extraction, exemplified by the dual U-Net model for thyroid nodule segmentation,[7] the cascaded U-Net for brain tumor substructure segmentation,[8] and the CU-Net framework for liver tumor segmentation.[9] However, such methods often result in a substantial increase in parameter count and computational cost. To address this limitation, Qin et al. proposed U^2-Net,[10] which utilizes a nested *U*-structure to achieve efficient multiscale feature fusion while controlling computational overhead. Building upon the U^2-Net architecture, ARes-U^{2}Net incorporates generative adversarial training, comprising a segmentation network and a discriminative network, to capture finer details and has demonstrated competitive performance in COVID-19 lesion segmentation. However, ARes-U^{2}Net adopts a two-stage pipeline: it first employs a separate U-Net to presegment lung parenchyma and then performs lesion segmentation on the masked result. This serial design not only increases computational complexity and processing time but also risks losing lesion regions located at the lung periphery.

Beyond *U*-shaped architectures, alternative models such as RefineNet[11] and PSPNet[12] enhance global context perception through multipath refinement and pyramid pooling, but they lack dedicated mechanisms for delineating blurred lesion boundaries and often require high computational resources, making them suboptimal for small and ambiguous COVID-19 lesions. 3D segmentation models such as V-Net[13] capture volumetric information, yet their heavy computational cost hinders practical deployment; our work therefore concentrates on efficient 2D processing while preserving multiscale context.

Generative adversarial networks (GANs) have been extensively explored for segmentation. Early GAN-based approaches introduced discriminators to refine mask quality,[14,15] but they often suffer from unstable gradient feedback and single-label outputs, limiting their suitability for dense medical image prediction. SegAN[16] and AlphaGAN[17] improved boundary sharpness via multiscale L_1 loss and adversarial training, yet these methods neither explicitly suppress interference from nonlung regions nor handle the severe class imbalance of small lesions—critical issues in COVID-19 CT scans. Our DDARes-U^{2}Net addresses these gaps by integrating adversarial training with a dual-decoder framework, in which a lung parenchyma decoder provides anatomical context to reduce false positives and an edge-aware loss sharpens lesion boundaries.

Recent GAN-based COVID-19 segmentation methods have introduced attention mechanisms and semisupervised learning to cope with complex lesion morphology and limited

annotations.[18–20] However, they typically do not jointly optimize lung region priors and boundary detail, which leaves them vulnerable to extraneous tissue artifacts and blurred margins. Although transfer learning has been widely adopted to mitigate the scarcity of annotated data, our approach tackles the same challenge through a task-specific dual-decoder design and a comprehensive multiscale loss, without dependence on external pretraining.

A landmark development in parallel to the above is the emergence of medical foundation models, most notably MedSAM.[21] By training on over 1.5 million image-mask pairs, MedSAM achieves remarkable cross-modal generalization and represents the state of the art in universal segmentation. MedSAM and the present work are best understood as complementary rather than competing: MedSAM optimizes for broad, zero-shot versatility, whereas our DDARes-U^{2}Net is a specialist model designed specifically for the tightly coupled challenges of COVID-19 CT segmentation—blurred boundaries, severe class imbalance, and nonlung interference—through a dedicated dual-decoder, edge-aware loss, and adversarial training. A direct performance comparison would therefore conflate different design objectives. We view these two paradigms as synergistic, and discussions on integrating our specialist components into foundation-model fine-tuning appear in the Future Work section.

The above analysis reveals several unmet needs in COVID-19 lesion segmentation: efficient multiscale feature fusion without excessive parameters, explicit suppression of nonlung interference, enhanced sensitivity to blurred boundaries, and robustness to class imbalance. Motivated by the comprehensive analysis of existing limitations in COVID-19 lesion segmentation, this paper revisits the encoder-dual decoder architecture and proposes DDARes-U^{2}Net, a novel framework specifically designed for accurate detection and segmentation of COVID-19 pneumonia lesions. The principal contributions of this work are threefold:

1. This paper proposes an innovative parallel segmentation model based on an encoder-dual decoder architecture that simultaneously accomplishes lung parenchyma segmentation and infected area detection within a unified framework. This architectural design eliminates the cumulative errors and missed lesion detection inherent in traditional serial presegmentation pipelines, thereby significantly improving both segmentation efficiency and completeness. The parallel processing approach ensures that both tasks benefit from shared feature representations while maintaining task-specific optimization objectives.
2. This paper designs an effective feature-sharing mechanism that facilitates synergistic collaboration between the two segmentation tasks. A unified encoder extracts discriminative high-level features from input images, which are then processed by two task-specific decoders. The lung parenchyma segmentation branch provides essential spatial prior knowledge to guide the overall segmentation process, effectively suppressing interference from nonlung regions and reducing false positive detections. Simultaneously, the COVID-19 lesion segmentation branch leverages these enhanced features for more accurate identification and delineation of infected areas, benefiting from the anatomical context provided by the parenchyma segmentation task.
3. This paper constructs a novel multiscale weighted loss function that incorporates explicit edge constraints to address the challenging problem of blurred lesion boundaries. By strengthening optimization penalties around lesion boundaries through edge-aware loss components, this function effectively mitigates segmentation inaccuracies caused by the indistinct margins characteristic of COVID-19 lesions. The edge-constrained loss mechanism significantly enhances the model's sensitivity to boundary regions and improves overall segmentation accuracy, particularly for lesions with ambiguous boundaries or complex shapes.

2 Research Background and Motivation

2.1 Analysis of the Research Problem

COVID-19 pneumonia lesions exhibit substantial variations in size within individual CT images, in which large consolidations and small ground-glass opacities frequently coexist. Their indistinct boundaries and irregular morphologies make accurate localization particularly challenging. Fine-grained regional details and edge information are susceptible to loss during the segmentation prediction process. Moreover, lesions typically occupy only a small fraction of the entire

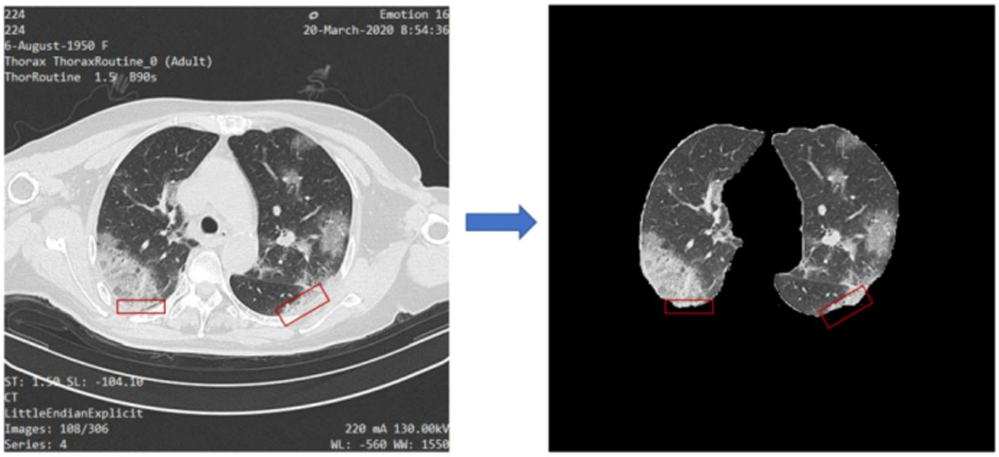


Fig. 1 Errors generated by U-Net in lung parenchyma segmentation.

image volume, leading to severe pixel-level class imbalance that complicates model training and convergence.

Current state-of-the-art methods such as ARes-U^{2}Net employ presegmented lung parenchyma images as input, requiring a preprocessing stage where original CT images undergo segmentation using a U-Net to obtain lung masks. Consequently, ARes-U^{2}Net operates as two-stage segmentation algorithms, which considerably increases computational complexity and processing time. More critically, the initial lung segmentation stage may result in loss of lesion regions. This problem arises because lesions are often distributed along the periphery of lung parenchyma, manifesting as gray-white ground-glass opacities. By contrast, healthy lung tissue typically appears as dark black regions with scattered vascular and tracheal shadows, whereas nonlung tissues outside the parenchyma present as clear-edged gray-white structures. When lesions are located at the lung boundary adjacent to other tissues, the lung segmentation process may misclassify portions of the lesion as nonlung tissue, introducing significant segmentation errors. As illustrated by the red bounding box in Fig. 1, such errors caused by lung parenchyma segmentation in the ARes-U^{2}Net preprocessing stage lead to partial loss of lesion areas adjacent to the lung periphery. This critical limitation of the two-stage design further motivates the development of our end-to-end DDARes-U^{2}Net framework.

2.2 Main Ideas

This paper proposes an end-to-end segmentation algorithm that eliminates the need for a two-stage process by enabling simultaneous segmentation of both lung parenchyma and lesions directly from raw CT images. The proposed model, termed Dual-Decoder Adversarial Residual U^{2}Net (DDARes-U^{2}Net), incorporates a dual-decoder architecture and an edge-aware loss function specifically tailored for COVID-19 lesion segmentation.

The encoder-dual decoder structure is designed to guide the algorithm to focus on lung parenchyma regions, thereby reducing interference from irrelevant anatomical structures. This approach not only mitigates the information loss inherent in preliminary lung segmentation but also shortens the overall processing time. Furthermore, DDARes-U^{2}Net is trained using a multiscale weighted loss function integrated with edge loss, enhancing the model's attention to the boundaries of COVID-19 lesions and improving segmentation accuracy for indistinct lesion margins.

3 DDARes-U^{2}Net

3.1 Model Framework

The DDARes-U^{2}Net algorithm utilizes an encoder-dual decoder architecture designed to simultaneously segment COVID-19 lesions and lung parenchyma. Both segmentation tasks share the same encoder and intermediate modules while employing separate decoders optimized for each specific task. This design guides the training process to focus on the lung parenchyma region,

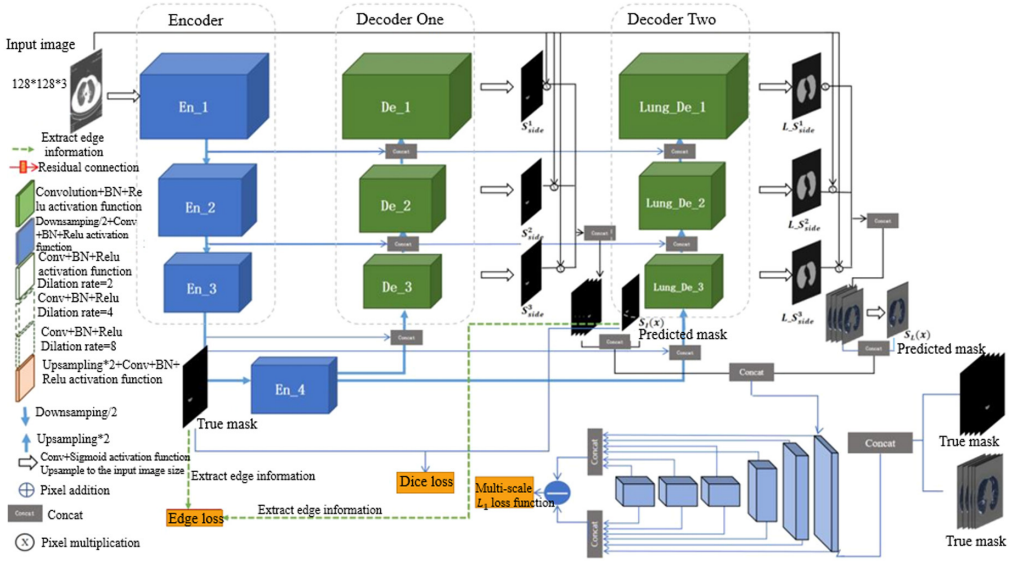


Fig. 2 DDARes-U²Net architecture.

thereby accurately identifying COVID-19 infected areas while reducing confusion and interference from other tissues.

The overall structure of DDARes-U²Net consists of two main components: a segmentation network and a discriminative network. The segmentation network follows an encoder-dual decoder architecture, in which the encoder takes the original CT image without presegmented lung parenchyma as input. As illustrated in Fig. 2, the left part depicts the segmentation network. Here, En_1 to En_3 represent the encoder modules, En_4 is the intermediate module, De_1 to De_3 form the first decoder, and Lung_De_1 to Lung_De_3 constitute the second decoder. Their internal designs are detailed in Fig. 3.

The output of the first decoder consists of three side outputs (S^1_{side}, S^2_{side}, S^3_{side}) from COVID-19 lesion segmentation. Each is subjected to element-wise multiplication with the input image and then concatenated along the channel dimension with the final segmentation result $S_L(x)$. Similarly, the output of the second decoder includes three side outputs ($L_S^1_{\text{side}}$, $L_S^2_{\text{side}}$, $L_S^3_{\text{side}}$) from lung parenchyma segmentation. These are also multiplied element-wise with the

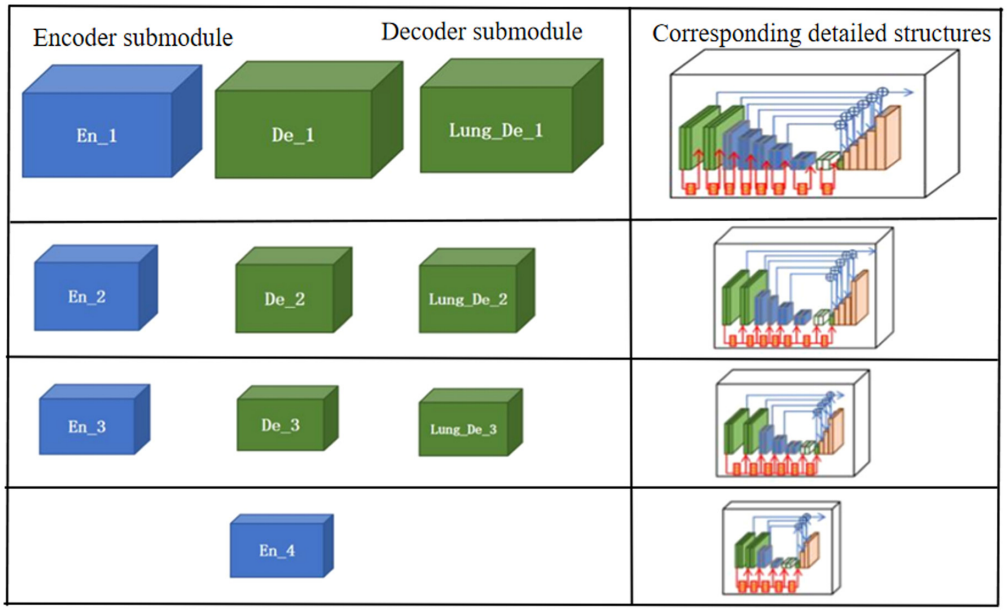


Fig. 3 DDARes-U²Net submodule structure.

input image and concatenated channel-wise with the final lung segmentation result $S_L(x)$. The overall output of the segmentation network is formed by concatenating the outputs of the two decoders along the channel dimension.

This combined output is fed into the discriminative network as predicted data. In addition, the real data input to the discriminative network is constructed as follows: the ground-truth COVID-19 lesion mask is multiplied element-wise with the input image, replicated 5 times along the channel dimension, and then concatenated with similarly processed lung parenchyma ground-truth mask (also multiplied element-wise with the input image and channel-replicated 5 times).

3.2 Segmentation Network

The segmentation network of DDARes-U²Net consists of a shared encoder and two structurally identical decoders, both symmetrically aligned with the encoder architecture. The detailed structure of each submodule is illustrated in Fig. 3. The submodules in DDARes-U²Net adopt a Res-UBlock design. Compared with the segmentation network of ARes-U²Net, the proposed network is two layers shallower.

Feature maps output from each submodule of the encoder are propagated through skip connections and concatenated with the upsampled feature maps from the corresponding higher-level submodules in both decoders. These concatenated features then serve as the input feature maps for the respective submodules in the decoder.

Overall, the input to the segmentation network is the original CT scan without lung parenchyma presegmentation. The output includes both the COVID-19 lesion segmentation mask and the lung parenchyma segmentation mask, along with their six corresponding side outputs. These outputs are fed into the discriminative network for evaluation using a multiscale L_1 loss. In addition, the COVID-19 lesion segmentation mask is further used to compute the Dice loss and the edge loss.

3.3 Loss Function

To address the limitations of the multiscale L_1 loss in segmenting COVID-19 lesion boundaries, DDARes-U²Net employs a comprehensive multiscale loss function with particular emphasis on infection edge delineation. Following the computation of adversarial loss via the multiscale L_1 loss (L_{mul}), an additional Dice loss (L_{dice}) is calculated between the predicted COVID-19 lesion segmentation mask and the ground truth. This encourages the model to prioritize the COVID-19 lesion segmentation task—the primary objective—over lung parenchyma segmentation.

For edge loss computation, morphological gradient operations are first applied to extract boundaries from both the ground truth and predicted masks. The edge loss (L_{edge}) is then computed using the Dice loss function between these extracted edge maps. The Dice loss is chosen over alternatives such as cross-entropy due to its superior performance in class-imbalanced segmentation tasks, as it is particularly suitable for scenarios with limited positive pixels. Edge regions—whether belonging to large or small lesions—inherently contain very few positive pixels. Finally, a comprehensive loss function (L_S) for training the segmentation network is formed by a weighted combination of the aforementioned losses, defined as follows:

$$\min_{\theta_S} L_S = \alpha L_{\text{mul}} + \beta L_{\text{dice}} + \gamma L_{\text{edge}}, \tag{1}$$

where $\alpha \leq \beta < \gamma$. The weight of L_{dice} is set slightly higher than that of L_{mul} to prioritize the COVID-19 lesion segmentation task over lung parenchyma segmentation. The highest weight is assigned to L_{edge} to emphasize accurate edge segmentation and enhance the detection of small lesions, preventing them from being overlooked during prediction. By minimizing the comprehensive loss function L_S, the segmentation network progressively improves both COVID-19 lesion and lung parenchyma segmentation results toward the ground truth masks, with enhanced focus on COVID-19 lesion segmentation and precise delineation of lesion boundaries.

The loss function for the discriminative network is defined as:

$$\max_{\theta_C} L_C = L_{\text{mul}}. \tag{2}$$

By maximizing the adversarial loss L_{mul}, the discriminative network enhances its ability to consistently and accurately distinguish real from generated data, thereby guiding the segmentation network toward precise segmentation of both COVID-19 lesions and lung parenchyma.

Given a dataset containing N training images x_n and their corresponding ground truth annotations y_n, the loss functions are defined as follows.

The loss L_{mul} [previously introduced in Eq. (2)] is given by

$$L_{\text{mul}} = \frac{1}{N}\sum_{n=1}^{N} \mathcal{L}_{\text{mea}}(f_C(A_n \circ S(x_n)), f_C(A_n \circ B_n)), \tag{3}$$

where

$$A_n = [x_n, x_n, x_n, x_n, x_n, x_n, x_n, x_n],$$

$$S(x_n) = [S^1_{\text{side}}(x_n), \cdots, S^3_{\text{side}}(x_n)], S_{\text{inf}}(x_n), \text{Lung_}S^1_{\text{side}}(x_n), \cdots, \text{Lung_}S^3_{\text{side}}(x_n), S_{\text{lung}}(x_n)],$$

$$B_n = [y_{\text{inf_}n}, y_{\text{inf_}n}, y_{\text{inf_}n}, y_{\text{inf_}n}, y_{\text{lung_}n}, y_{\text{lung_}n}, y_{\text{lung_}n}, y_{\text{lung_}n}],$$

$$\mathcal{L}_{\text{mea}}(f_C(x), f_C(x')) = \frac{1}{L}\sum_{i=1}^{L} \|f^i_C(x) - f^i_C(x')\|_1.$$

Here, N denotes the batch size; A_n represents the input image concatenated 8 times along the channel dimension; $\mathcal{L}_{\text{mea}}$ denotes the mean absolute error (MAE) computed via the L_1 distance; $S(x_n)$ refers to the channel-wise concatenation of the three side outputs and one final prediction from decoder 1 (for COVID-19 lesion segmentation) and the three side outputs and one final prediction from decoder 2 (for lung parenchyma segmentation); $A_n \circ S(x_n)$ denotes the eight-layer input image masked by the predicted masks (including six side outputs and two final predictions) from the two decoders of the segmentation network, i.e., the element-wise multiplication between A_n and the predicted mask $S(x_n)$; B_n is obtained by concatenating the ground truth COVID-19 lesion mask 4 times along the channel dimension and the ground truth lung parenchyma mask 4 times along the channel dimension, followed by concatenating these two sets together; $A_n \circ B_n$ represents the eight-layer input image masked by the eight-layer ground truth mask, i.e., the element-wise multiplication between A_n and B_n; and $f_C(x)$ denotes the multilevel features extracted by the discriminative network.

The Dice loss $\mathcal{L}_{\text{dice}}$ is defined as follows:

$$L_{\text{dice}} = 1 - \frac{2 * \sum_i^{W \cdot H} G_{R_i} * p_i}{\sum_i^{W \cdot H} G^2_{R_i} + \sum_i^{W \cdot H} p_i^2}. \tag{4}$$

Here, W and H denote the width and height of the segmentation mask, respectively. $G_{R_i} \in \{0, 1\}$ and p_i represent the ground truth label and predicted probability of the COVID-19 lesion for pixel i.

$$L_{\text{edge}} = -\frac{1}{N}\sum_{n=1}^{N} l_{\text{dice}}(f_{\text{edge}}(y_n), f_{\text{edge}}(S_{\text{inf}}(x_n))), \tag{5}$$

where

$$l_{\text{dice}} = 1 - \frac{2 * \sum_i^{W \cdot H} G_{E_i} * p_i}{\sum_i^{W \cdot H} G^2_{E_i} + \sum_i^{W \cdot H} p^2},$$

$$f_{\text{edge}}(X) = X \oplus K - X \ominus K,$$

$$X \oplus K = \{x | (K)_x \cap X \neq \varnothing\},$$

$$X \ominus K = \{x | K + x \subseteq X\},$$

where l_{dice} denotes the Dice loss. After extracting the edge pixels from both the ground truth mask and the predicted mask, the Dice loss is computed between the ground truth edge pixels and the predicted edge pixels. Here, W and H represent the width and height of the mask, respectively. $G_{E_i} \in \{0, 1\}$ and p_i denote the ground truth edge label and the predicted probability of pixel i, respectively. $f_{\text{edge}}(X)$ represents the morphological gradient function, which first

dilates the boundary of X using $X \oplus K$ (dilation), then erodes it using $X \ominus K$ (erosion), $(K)_x = \{k + x | k \in K\}$ denotes the translation of the structuring element K by the vector x, and finally computes the difference between the two operations to preserve the edge contours of the COVID-19 lesions.

4 Experimental Design and Analysis

This section presents the comprehensive experimental framework, including dataset configurations, experimental setup, and detailed analysis of results. The dataset description encompasses both self-constructed datasets and publicly available benchmarks. The experimental setup covers the training environment, hyperparameter configurations, and evaluation metrics. The results and analysis section presents both comparative experiments and ablation studies. Comparative experiments are conducted under two settings: within-dataset evaluation and cross-dataset evaluation. Ablation studies are performed exclusively in the within-dataset setting to validate the contribution of each architectural component.

4.1 Datasets

Tissue damage caused by COVID-19 manifests as characteristic changes in medical imaging, making segmentation algorithms a viable tool for delineating infected lesions. Commonly used medical imaging modalities include computed tomography (CT), magnetic resonance imaging (MRI), X-ray imaging, and ultrasound. Among these, CT imaging employs ionizing radiation to scan cross-sections of the body at fine intervals, producing detailed images of internal structures. CT exhibits superior clarity in characterizing COVID-19 lesions compared with other imaging modalities.[22]

Relative to diagnostic methods such as chest X-ray and polymerase chain reaction (PCR) testing, chest CT scans offer higher sensitivity in detecting and localizing COVID-19 infections. This is particularly valuable in early or asymptomatic cases where CT may reveal lung abnormalities even before PCR returns positive results, enabling earlier diagnosis and isolation. CT imaging provides richer visual information and higher spatial resolution, allowing detailed assessment of lesion morphology, size, density, and distribution. Based on these characteristics, chest CT scans can be used to stage disease severity and monitor progression by evaluating the number, size, and distribution of pulmonary lesions. The quantitative analysis enabled by CT imaging provides objective metrics for clinical decision-making and treatment planning.

Due to privacy restrictions on patient data and the resource-intensive nature of lesion annotation—requiring significant time and expertise from experienced radiologists—publicly available CT image datasets for COVID-19 segmentation remain limited in scale.[23–25] As a result, most current studies train their algorithms using relatively small datasets,[26,27] with some even employing only infected slices for both training and testing.[28–30] However, in clinical practice, infections typically appear only in a subset of slices within a CT scan sequence rather than throughout the entire volume. To better simulate real clinical scenarios and enhance model generalization, this study incorporates chest CT slices from both infected and noninfected COVID-19 patients for training and evaluating the segmentation method. To enable a comprehensive assessment of algorithmic performance in COVID-19 lesion segmentation, we employ one self-constructed dataset and four public datasets[31–33] as the experimental foundation. The evaluation framework includes both within-dataset and cross-dataset experiments, ensuring a thorough and robust analysis of segmentation performance across diverse data sources. The following sections detail the self-built and public datasets used in this study.

4.1.1 *Self-constructed dataset*

This study developed a specialized COVID-19 dataset for lesion segmentation tasks, containing 2412 high-resolution CT scan images sourced from 261 patients (mean age: 49 ± 15 years; 113 males, 148 females) treated at Beijing Ditan Hospital in 2021. All patient identifiers were permanently removed to comply with medical data privacy regulations. The raw data were provided in DICOM format. The preprocessing pipeline included:

1. The DICOM source files were first converted to 512×512-pixel PNG format images using DicomConverter software.

2. Radiologists performed manual annotation of the images using labelme software, generating corresponding JSON annotation files.
3. The JSON files were transformed into structured folders containing both lesion segmentation masks and original CT images.
4. A U-Net model for lung parenchyma segmentation was trained using deep learning techniques, producing lung parenchyma masks that enabled precise extraction of pulmonary regions.
5. Binarization was applied to the lesion egmentation masks to create binary mask images.
6. Final images underwent preprocessing including flipping and normalization before model input, resulting in standardized 128×128-pixel input samples.

The COVID-19 dataset exhibits significant variability in pneumonia lesion severity across individual CT scans. Lesion pixel ratios in single images range from 1.02% to 38.58% of total pixels, with a mean value of 4.02%. For model development, 1688 slices were selected from 210 scan sequences as training data, whereas 724 slices from 51 scan sequences constituted the test set, following a 7:3 training-testing split. Each CT slice contains nonpulmonary artifacts beyond the lung parenchyma regions, including patient torso structures, scanner table components, textual annotations, and extraneous anatomical structures (as shown in Fig. 4).

To mitigate the impact of extraneous artifacts in chest CT scans on algorithm performance and enhance focus on COVID-19 lesion regions within the lung parenchyma, this study implemented a U-Net–based lung parenchyma segmentation framework. The U-Net architecture adopted for this task maintains structural simplicity, as chest CT imaging demonstrates clear anatomical boundaries between lung parenchyma and adjacent thoracic structures, with positively annotated pixels constituting a significant proportion of the total image content. This enables the classical U-Net architecture to achieve effective parenchyma segmentation without requiring deeper or more complex variants. Figure 5 illustrates the U-Net architecture employed for lung parenchyma segmentation, whereas Fig. 6 provides qualitative comparisons demonstrating segmentation efficacy through pre- and postprocessing visualizations. To validate the reliability of this lung parenchyma segmentation model, we evaluated its performance on the test set of dataset I: Dice = 0.9723, IoU = 0.9461, Specificity = 0.9989, and Precision = 0.9756. Furthermore, an error analysis on 100 randomly selected test slices showed that only 1.2% of slices exhibited any lesion region loss attributable to parenchyma segmentation errors, confirming the minimal impact of the presegmentation step.

4.1.2 *Public datasets*

To comprehensively evaluate algorithm performance, this study incorporated both the self-constructed dataset and four established public datasets in experimental validation:

1. COVID-19 CT Segmentation Dataset[31] comprises 100 annotated CT slices from over 40 COVID-19 patients.
2. Segmentation Dataset Nr.2[31] contains 9 CT scan sequences (829 total slices), with 373 slices annotated for COVID-19 infection regions. The remaining slices exhibit no COVID-19 pathology.

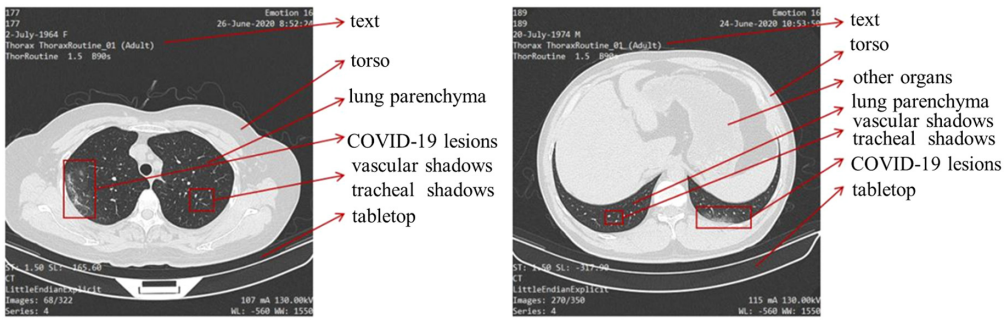


Fig. 4 Information contained in patient's chest CT scan images.

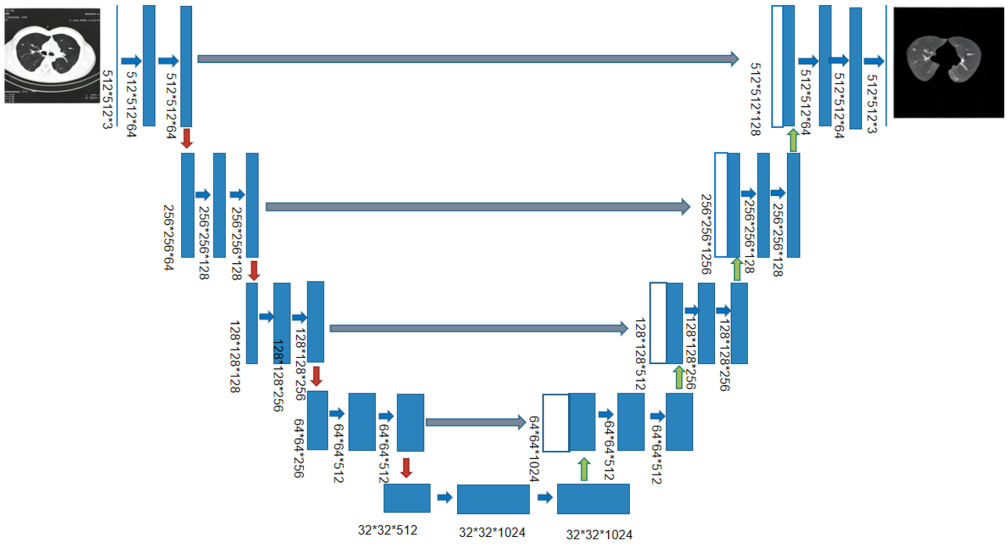


Fig. 5 U-Net algorithm architecture for lung parenchyma segmentation.

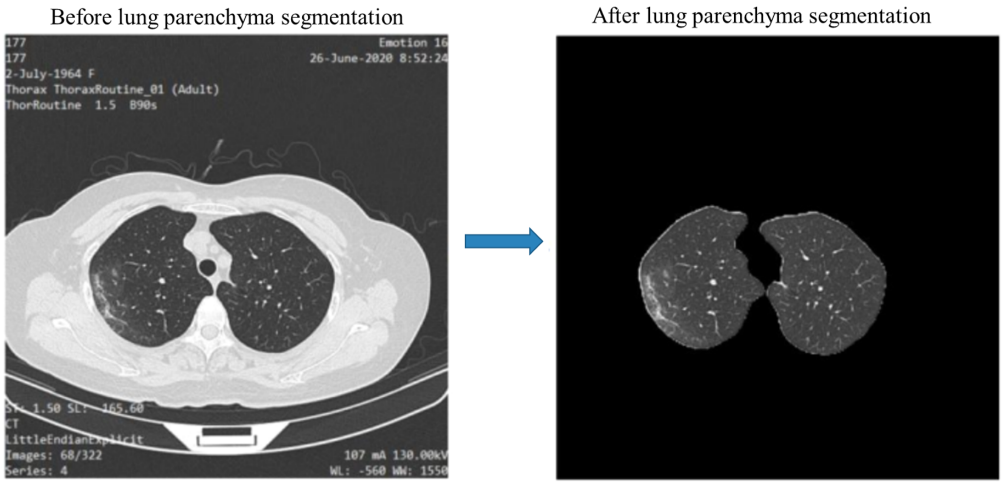


Fig. 6 Comparison of lung parenchyma segmentation before and after.

3. COVID-19-CT-Seg Dataset[32] includes 20 CT scan sequences with annotated COVID-19 regions, comprising 3520 total slices (1844 infection-positive).
4. The CC-CCII Segmentation Dataset[33] comprises 750 CT slices from 150 COVID-19 patients, with manual annotations identifying the background, lung lobes, ground-glass opacity (GGO), and consolidation.

For simplicity, the self-constructed dataset in this study is referred to as dataset I, whereas the other four publicly available datasets are designated as datasets II to V; all these datasets include ground truth masks for lung parenchyma segmentation. The detailed characteristics of these datasets are presented in Table 1.

4.2 Experimental Setup

The experiments in this study were implemented using the PyTorch framework with an NVIDIA GeForce RTX 3090 GPU (24 GB VRAM). All convolutional layer weights were initialized using a Gaussian distribution with a mean of 1 and a variance of $\sqrt{\frac{2}{(W_K * H_K * C_{\text{out}})}}$, where W_K and H_K represent the width and height of the convolutional kernel, respectively, and C_{out} denotes the number of output channels. Biases were initialized to zero. The Adam optimizer was employed

Table 1 Introduction of the dataset.

Dataset name	Source	Number of CT sequences	Number of slices
Dataset I	Self-constructed dataset (Beijing Ditan Hospital)	261	2188
Dataset II	COVID-19 CT segmentation dataset	40	100
Dataset III	Segmentation dataset Nr.2	9	829
Dataset IV	COVID-19-CT-Seg dataset	20	3520
Dataset V	CC-CCII segmentation dataset	150	750

to train all models, with β_1 and β_2 set to 0.9 and 0.999, respectively. The training was conducted with a batch size of 6 for 60 epochs. The initial learning rate was set to 0.01, which was reduced by a factor of 0.1 after 30 epochs and again after 50 epochs.

The evaluation metrics used in this experiment include the Dice coefficient (Dice score), intersection over union (IoU), specificity (Spec), and precision (Prec). The corresponding calculation formulas are as follows:

$$\text{Dice} = \frac{1}{N}\sum_{i=1}^{N}\frac{2\text{TP}_i}{2\text{TP}_i + \text{FP}_i + \text{FN}},$$

$$\text{IoU} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}},$$

$$\text{spec} = \frac{\text{TN}}{\text{FP} + \text{TN}},$$

$$\text{prec} = \frac{\text{TP}}{\text{TP} + \text{FP}},$$

where N denotes the number of images in the test set, TP (true positive) represents the number of pixels where the model correctly predicts positive cases as positive, FP (false positive) indicates the number of pixels where the model incorrectly predicts negative cases as positive, TN (true negative) refers to the number of pixels where the model correctly predicts negative cases as negative, and FN (false negative) signifies the number of pixels where the model incorrectly predicts positive cases as negative.

This section designs two types of experiments: intradataset experiments and cross-dataset experiments. Intradataset experiments refer to scenarios, where both the training and test sets are derived from the same dataset. Cross-dataset experiments involve training and test sets originating from different datasets. The algorithms used for comparison include six representative models: PDEAtt-UNet, Att-UNet, UNet++, SCOATNet, and ARes-U^{2}Net. Detailed descriptions of these six algorithms are provided below:

1. PDEAtt-UNet[34]: A deep learning algorithm for COVID-19 lesion segmentation from CT scans, serving as a comparative algorithm within the same domain. It employs a pyramid dual-decoder architecture and captures global contextual awareness from all encoder layers through multiple attention gates.
2. Att-UNet[4]: A classical medical image segmentation algorithm used as a baseline comparative model. It integrates three types of attention gate structures, which automatically suppress irrelevant features while highlighting task-relevant characteristics.
3. UNet++[6]: A classic medical image segmentation algorithm designed with a densely connected architecture, adopted as a baseline comparative method. It redefines skip connections to enable multiscale feature fusion, thereby reducing the semantic gap between features at different scales.
4. SCOATNet[19]: A deep learning algorithm for COVID-19 lesion segmentation from CT scans, included as a same-domain comparative model. Based on the UNet++ architecture,

it incorporates spatial and channel attention modules to extract pixel-wise and channel-wise salient features of infected regions.

5. ARes-U²Net[35]: The algorithm draws inspiration from the idea of generative adversarial algorithms and comprises a segmentation network and a discriminative network. The adversarial training process enables the segmentation model to identify more detailed features.

Collectively, the five baseline methods introduced above and our proposed model form six approaches for comparison, which can be categorized into three groups corresponding to the literature reviewed in the Introduction.

1. **U-Net—based architectures:** Att-UNet and UNet++, representative of the encoder–decoder paradigms discussed in Sec. 1, paragraph 1.
2. **GAN-based models:** ARes-U²Net, which employs adversarial training and echoes the GAN-based models reviewed in Sec. 1, paragraph 3.
3. **Task-specific COVID-19 segmentation models:** PDEAtt-UNet and SCOATNet, which are dedicated to COVID-19 lesion segmentation and correspond to the task-oriented methods surveyed in Sec. 1, paragraph 4.

4.3 Intra-Dataset Comparative Experiments

Table 2 presents the comprehensive results of intradataset comparative experiments conducted on dataset I and IV using different algorithms, with the performance of DDARes-U²Net highlighted in bold. The corresponding bar charts are illustrated in Figs. 7 and 8. Experimental results

Table 2 Comparative experiments within dataset. Bold values represent the best performance.

	Dataset I				Dataset IV			
	Dice	IoU	Spec	Prec	Dice	IoU	Spec	Prec
SCOATNet	0.5529	0.5021	0.9958	0.6729	0.3674	0.5506	0.9984	0.8429
UNet++	0.5488	0.5186	0.9960	0.6631	0.3743	0.5372	0.9982	0.8239
Att-UNet	0.5599	0.5266	0.9962	0.6722	0.3342	0.5229	0.9984	0.8345
PDEAtt-UNet	0.5631	0.5323	0.9962	0.6671	0.4128	0.6067	0.9952	0.7003
ARes-U²Net	0.6499	0.5459	0.9977	0.7673	0.4468	0.7617	0.9982	0.8452
DDARes-U²Net	**0.6534**	**0.5465**	**0.9980**	**0.7685**	**0.4556**	**0.7887**	**0.9985**	**0.8810**

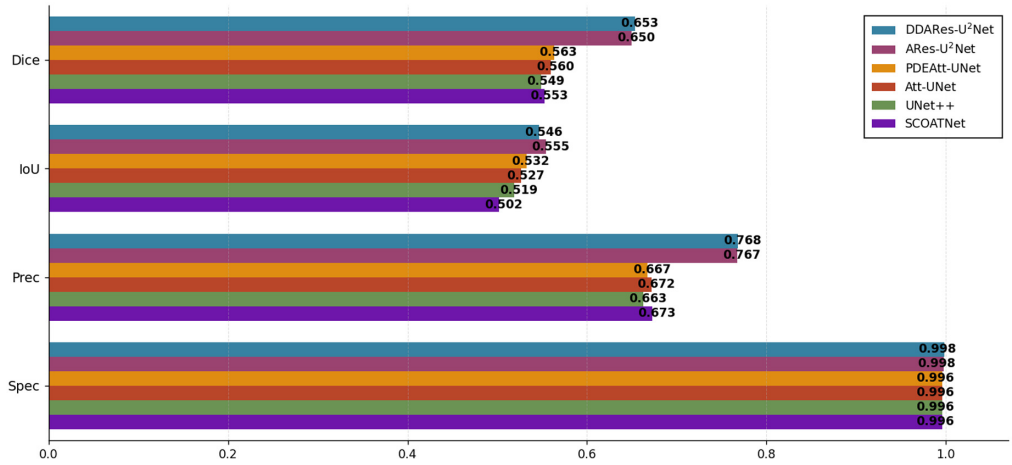


Fig. 7 Bar chart of comparative experiments within dataset I.

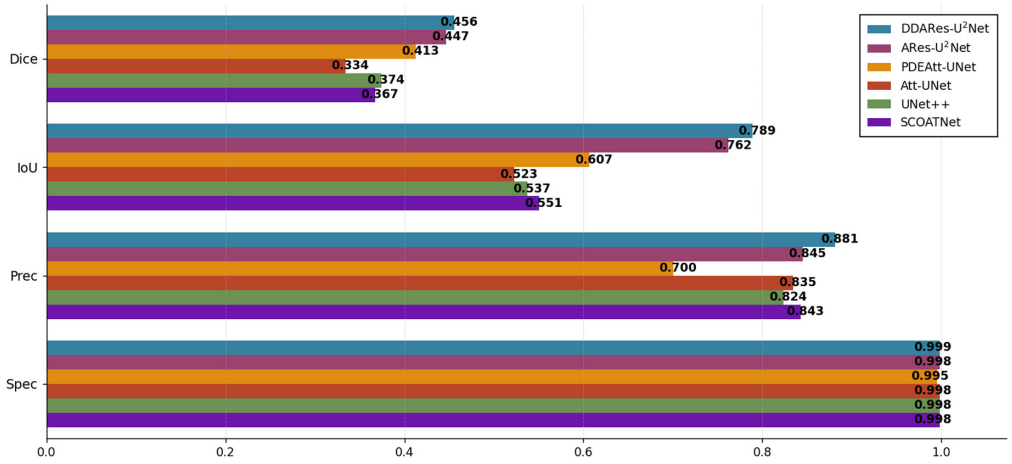


Fig. 8 Bar chart of comparative experiments within dataset IV.

demonstrate that DDARes-U^{2}Net achieves superior performance across multiple evaluation metrics compared with existing state-of-the-art algorithms. Specifically, in dataset I, DDARes-U^{2}Net attains Dice and IoU scores of 65.34% and 54.65%, respectively. The strongest competing algorithm, ARes-U^{2}Net, achieves values of 64.99% and 54.59% for these metrics, indicating that DDARes-U^{2}Net outperforms it by margins of 0.35% and 0.06%. In dataset IV, DDARes-U^{2}Net demonstrates even more substantial improvements, surpassing all competing architectures. Notably, it improves Dice and IoU scores by 0.88% and 2.70%, respectively, compared with ARes-U^{2}Net.

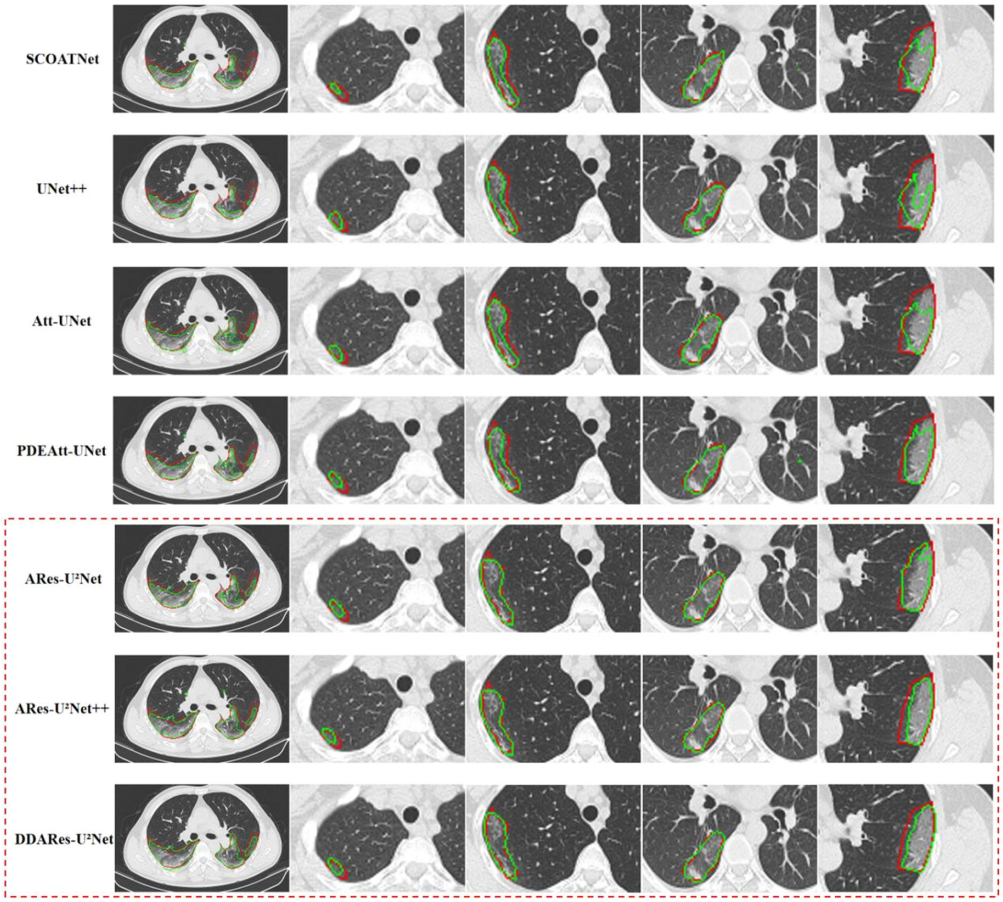


Fig. 9 Visualization of comparative experiments within dataset I (red contours: ground truth; green contours: model predictions).

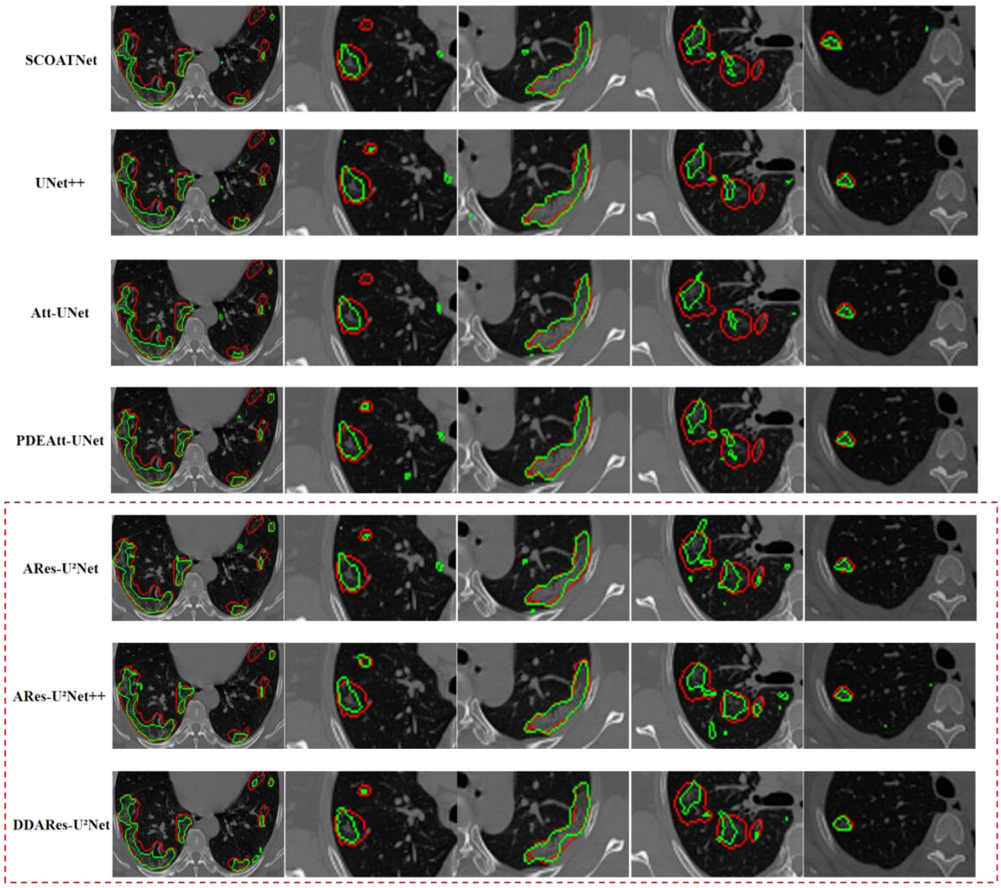


Fig. 10 Visualization of comparative experiments within dataset IV (red contours: ground truth; green contours: model predictions).

As illustrated in Figs. 9 and 10, the qualitative visualization results of the intradataset comparative experiments are presented. The regions enclosed by red contours represent ground truth lesion boundaries, whereas the areas enclosed by green contours indicate lesion boundaries predicted by various comparative algorithms. These visualizations provide clear evidence of the superior segmentation accuracy achieved by DDARes-U^{2}Net, particularly in delineating complex lesion boundaries and detecting small lesions that may be overlooked by other methods.

Figure 9 displays the qualitative visualization results of the intradataset experiment conducted on dataset I. Among the compared algorithms, ARes-U^{2}Net and DDARes-U^{2}Net demonstrate superior performance in terms of COVID-19 lesion segmentation completeness. Specifically, ARes-U^{2}Net achieves relatively complete lesion segmentation with accurate spatial localization. ARes-U^{2}Net exhibits enhanced segmentation quality compared with its predecessor, whereas DDARes-U^{2}Net excels in precise boundary delineation and delivers the most accurate segmentation results overall.

Figure 10 presents the visualization results of the intradataset experiment on dataset IV. Despite the relatively poor image quality and blurred boundaries characteristic of CT scans in dataset IV, DDARes-U^{2}Net consistently achieves the best segmentation performance, demonstrating robustness to image degradation.

4.4 Cross-Dataset Comparative Experiments

To comprehensively validate the generalization capability of DDARes-U^{2}Net, this study designed extensive cross-dataset experiments where models were trained on dataset I, III, and IV and tested under challenging cross-dataset scenarios. Tables 3 and 4 present the results of training on dataset I and testing on datasets III to V. The results demonstrate modest performance

Table 3 Cross-dataset comparative experiments (trained on dataset I, tested on dataset II–III). Bold values represent the best performance.

	Testing: dataset II				Testing: dataset III			
Model	Dice	IoU	Spec	Prec	Dice	IoU	Spec	Prec
SCOATNet	0.4334	0.3991	0.9678	0.6211	0.1633	0.3354	0.9705	0.4520
UNet++	0.5488	0.5786	0.9760	0.6614	0.2180	0.4349	0.9921	0.4446
Att-UNet	0.4453	0.4151	0.9748	0.6116	0.2177	0.4523	0.9956	0.5454
PDEAtt-UNet	0.5106	0.4787	0.9673	0.6131	0.2153	0.4688	0.9948	0.5569
ARes-U^{2}Net	0.5489	0.5892	0.9765	0.6713	0.2928	0.5160	0.9984	0.5720
DDARes–U^{2}Net	**0.5965**	**0.5984**	**0.9783**	**0.6842**	**0.2985**	**0.5430**	**0.9986**	**0.5871**

Table 4 Cross-dataset comparative experiments (training on dataset I, testing on datasets IV to V). Bold values represent the best performance.

	Testing: dataset IV				Testing: dataset V			
Model	Dice	IoU	Spec	Prec	Dice	IoU	Spec	Prec
SCOATNet	0.1857	0.3691	0.9913	0.5074	0.3662	0.4131	0.9914	0.5280
UNet++	0.2525	0.4686	0.9986	0.6111	0.3699	0.4469	0.9930	0.5429
Att-UNet	0.2219	0.4470	0.9935	0.5135	0.3745	0.4563	0.9925	0.5359
PDEAtt-UNet	0.2714	0.4124	0.9977	0.5478	0.3808	0.4644	0.9919	0.5230
ARes-U^{2}Net	0.2964	0.4782	0.9988	0.6941	0.4386	0.4707	0.9961	0.5471
DDARes–U^{2}Net	**0.3092**	**0.5094**	**0.9989**	**0.6973**	**0.4701**	**0.5194**	**0.9972**	**0.5761**

improvements when testing on datasets III and IV, and more substantial improvements when testing on datasets II and V.

Table 5 presents the results of models trained on dataset III and tested on datasets II, IV, and V, respectively. Table 6 shows the results of models trained on dataset IV and tested on datasets II, III, and V. Among these challenging scenarios, DDARes-U^{2}Net demonstrates the most

Table 5 Cross-dataset comparative experiments (training on dataset III, testing on datasets II, IV, and V). Bold values represent the best performance.

	Testing: dataset II		Testing: dataset IV		Testing: dataset V	
Model	Dice	IoU	Dice	IoU	Dice	IoU
SCOATNet	0.4082	0.3266	0.2736	0.4343	0.2140	0.2691
UNet++	0.2479	0.1533	0.2303	0.3508	0.1877	0.2058
Att-UNet	0.4101	0.3034	0.2590	0.3948	0.1724	0.2029
PDEAtt-UNet	0.4613	0.4035	0.2945	0.4708	0.2439	0.3045
ARes-U^{2}Net	0.5744	0.4304	0.3033	0.4759	0.2997	0.3050
DDARes–U^{2}Net	**0.6072**	**0.4735**	**0.3345**	**0.4812**	**0.3034**	**0.3300**

Table 6 Cross-dataset comparative experiments (training on dataset IV, testing on datasets II, III, and V). Bold values represent the best performance.

	Testing: dataset II		Testing: dataset III		Testing: dataset V	
Model	Dice	IoU	Dice	IoU	Dice	IoU
SCOATNet	0.5952	0.4637	0.1823	0.2050	0.3422	0.4681
UNet++	0.5843	0.5088	0.2695	0.5420	0.1684	0.2465
Att-UNet	0.3837	0.2899	0.2510	0.3839	0.2378	0.3617
PDEAtt-UNet	0.6318	0.5383	0.2796	0.5643	0.3890	0.5008
ARes-U²Net	0.6447	0.5413	0.2849	0.5879	0.3929	0.5242
DDARes-U²Net	**0.6962**	**0.5436**	**0.2892**	**0.5893**	**0.4059**	**0.5395**

significant performance improvement in Table 6 when trained on dataset IV and tested on dataset II, with Dice and IoU scores increasing by 5.51% and 0.23% compared with ARes-U²Net.

These comprehensive results demonstrate that DDARes-U²Net achieves superior performance across diverse evaluation scenarios, with particularly pronounced improvements observed in two specific cross-dataset configurations: when trained on dataset I and tested on datasets II and V, and when trained on dataset IV and tested on dataset II. These findings validate the robust generalization capability and practical clinical utility of the proposed method. Notably, DDARes-U²Net consistently outperforms all compared methods across every cross-dataset configuration in Tables 3–6, despite the substantial heterogeneity among the five independent cohorts in terms of scanning protocols, patient demographics, and lesion characteristics. The most pronounced gains emerge in the most challenging transfer scenarios—for instance, when trained on dataset IV and tested on dataset II, the model achieves a $+5.51$ pp Dice improvement over ARes-U²Net—demonstrating strong robustness to domain shift.

4.5 Ablation Study

In the ablation study, dataset I and dataset IV—which contain a larger number of CT scans—were selected for intradataset experiments. This study aims to elucidate the individual contributions and effects of key components in the DDARes-U²Net algorithm, thereby determining the optimal configuration for achieving the best segmentation performance. Four experimental configurations were established, each incorporating specific modules as detailed in Table 7:

1. Incorporates the Res-UBlock submodule into the base U²Net architecture.
2. Integrates both the Res-UBlock submodule and an encoder-dual-decoder design (four-layer structure) into the base U²Net.
3. Represents the proposed network architecture, which includes DDRes-U²Net as the segmentation network and a multiscale fusion discriminative module. Training employs a multiscale L_1 loss function for optimization.

Table 7 Comparison of different model components.

No.	Res-U²Net	Dual decoder	Discriminative network	Multiscale L_1 loss	Comprehensive loss function
A	✓				
B	✓	✓			
C	✓	✓	✓	✓	
D	✓	✓	✓		✓

Table 8 Intradataset ablation experiments.

	Dataset I				Dataset IV			
No.	Dice	IoU	Spec	Prec	Dice	IoU	Spec	Prec
A	0.6065	0.5020	0.9970	0.7340	0.4033	0.6100	0.9957	0.8406
B	0.6162	0.5231	0.9975	0.7497	0.4320	0.7110	0.9973	0.8420
C	0.6505	0.5463	0.9977	0.7685	0.4515	0.7742	0.9983	0.8706
D	0.6534	0.5465	0.9980	0.7685	0.4556	0.7887	0.9985	0.8810

4. Refines the proposed architecture (the same components as *C*) with an infection-prioritized multiscale comprehensive loss function during training to emphasize edge-aware segmentation performance.

Table 8 presents the results of the ablation study, together with the corresponding line charts in Figs. 11 and 12 and the visual comparisons in Fig. 13. Notably, the progressive improvement from configurations A to D is accompanied by a consistent reduction in false negative regions, as evidenced by the more complete coverage of ground-truth lesion areas in both the quantitative metrics (Dice, IoU) and the visual comparisons (Fig. 13). The edge-aware loss specifically targets indistinct boundaries—the primary source of missed detections—thereby directly addressing the clinical concern of lesion undersegmentation. From these results, we draw the following observations:

1. Configuration B outperforms A: On dataset I, B achieves improvements of 0.97% in Dice and 2.11% in IoU compared with A; on dataset IV, it shows gains of 2.87% in Dice and 10.10% in IoU. These results demonstrate that simultaneously segmenting COVID-19 lesions and lung parenchyma via the encoder-dual-decoder design contributes to higher segmentation accuracy compared with the encoder–decoder architecture.
2. Configuration C outperforms B: On dataset I, C achieves marginal improvements of 0.06% in Dice and 0.04% in IoU over B; on dataset IV, it exhibits gains of 0.47% in Dice and 1.25% in IoU. This indicates that replacing the segmentation network in ARes-U²Net with DDRes-U²Net (which concurrently segments lung parenchyma and COVID-19 lesions) and integrating a multiscale fusion discriminative module trained with multiscale L_1 loss further enhances segmentation precision.
3. Configuration D outperforms C: On dataset I, D improves Dice by 0.29% and IoU by 0.02% over C; on dataset IV, it achieves gains of 0.41% in Dice and 1.45% in IoU. These findings confirm that employing an infection-edge-focused multiscale comprehensive loss function

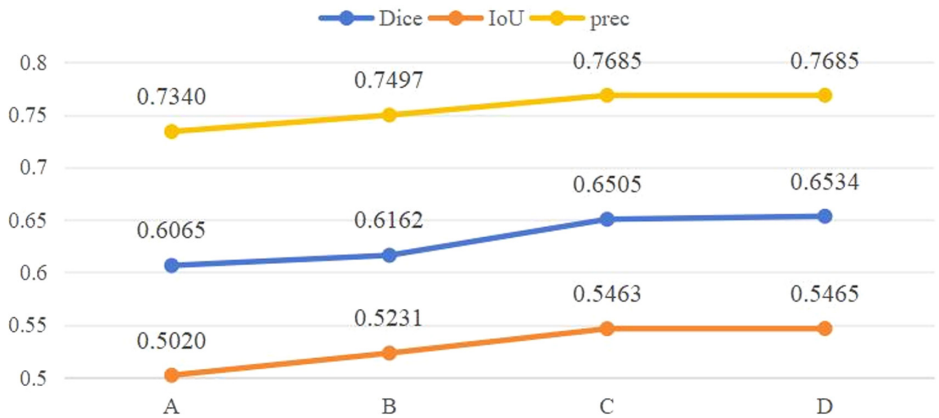


Fig. 11 Line chart of ablation experiments within dataset I.

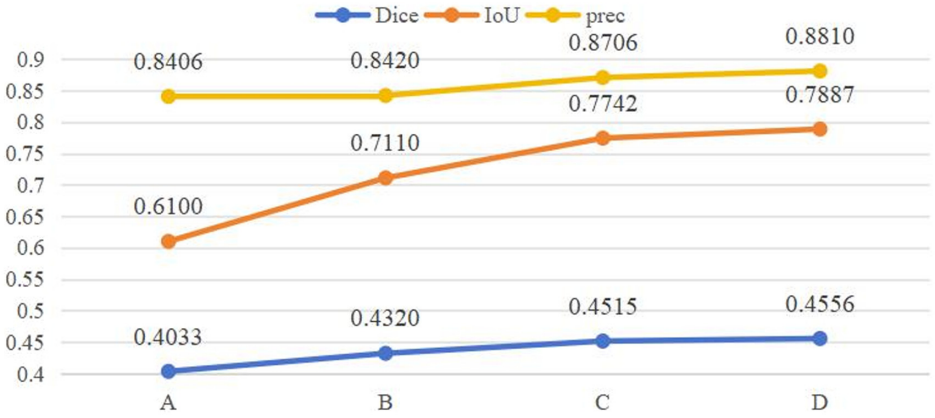


Fig. 12 Line chart o ablation experiments within dataset IV.

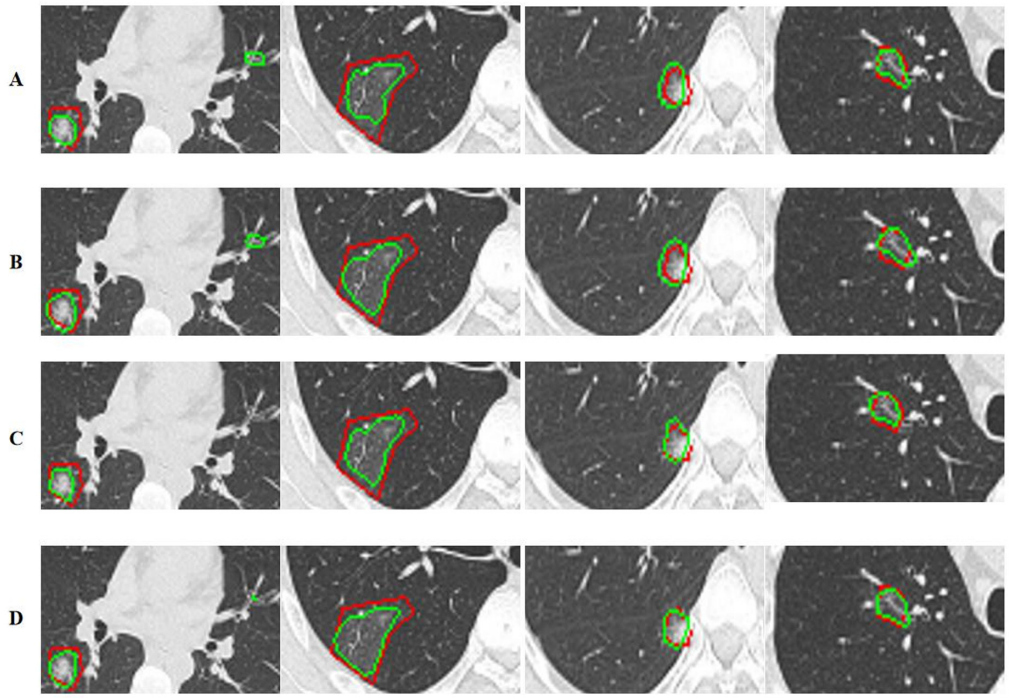


Fig. 13 Visualization of intradataset ablation experiments within dataset I (red contours: ground-truth; green contours: model predictions).

yields superior experimental outcomes compared with the multiscale L_1 loss function for COVID-19 lesion segmentation tasks.

As shown in Fig. 13, configuration B produces more complete and accurate COVID-19 lesion segmentation compared with A, validating the efficacy of the dual-decoder design in improving segmentation precision. Configuration C demonstrates reduced susceptibility to noise and a lower false-positive rate relative to B. Configuration D achieves the most precise COVID-19 lesion segmentation with minimal noise interference and notably outperforms C in capturing detailed lesion boundaries that align closely with ground-truth masks. These results confirm that the infection-edge-focused multiscale comprehensive loss function significantly enhances boundary segmentation accuracy. In this experiment, the segmentation network of DDARes-U²Net contains 30.74 M parameters, representing a substantial reduction compared with ARes-U²Net (49.09 M). This 37.4% reduction in model size, combined with the consistent accuracy improvements observed across configurations B to D (Table 8), demonstrates that our

encoder–dual-decoder design achieves a favorable efficiency–performance trade-off. Unlike competing methods that require computationally expensive components—such as the pyramid dual-decoder attention in PDEAtt-UNet, the nested dense connections in UNet++, or the two-stage adversarial pipeline in ARes-U^{2}Net—DDARes-U^{2}Net delivers competitive or superior segmentation accuracy within a lightweight, single-stage framework. This makes it substantially more suitable for deployment in resource-limited clinical settings where large models are impractical.

5 Conclusion

The DDARes-U^{2}Net model represents a significant advancement in COVID-19 lesion segmentation, introducing an innovative encoder-dual-decoder architecture that simultaneously addresses lung parenchyma and lesion segmentation through a unified framework. This paper has demonstrated how the integration of specialized architectural components—the dual-decoder structure and the multiscale comprehensive loss function—enables superior segmentation performance while maintaining computational efficiency. The model's achievement of 65.34% Dice score and 54.65% IoU score on dataset I, combined with its remarkable cross-dataset generalization capabilities, establishes its potential as a robust clinical tool. The efficiency of the model, with only 30.74 M parameters, represents a breakthrough in achieving high-accuracy medical image segmentation with reduced computational requirements. Moreover, its consistent superiority across all five independent CT cohorts—in both intra-dataset (Tables 2) and cross-dataset settings (Tables 3–6)—confirms the model's robust generalisability beyond single-center data. This efficiency, coupled with the elimination of preprocessing steps through its integrated dual-task approach, positions DDARes-U^{2}Net as a practical solution for real-world clinical deployment.

6 Future Work

This work demonstrates that the proposed DDARes-U^{2}Net achieves strong performance in COVID-19 lesion segmentation, although several avenues for future improvement remain.

1. **Extension to 3D segmentation.** The model could be extended to 3D segmentation by integrating 3D convolutional layers and volumetric feature extraction modules, thereby better capturing interslice spatial continuity and improving accuracy for large, irregular lesions.
2. **Semisupervised and weakly supervised learning.** To address the scarcity of annotated medical data, semisupervised and weakly supervised learning strategies can be explored, reducing reliance on manual annotations and enabling effective training with fewer labeled samples.
3. **Improved cross-scanner and cross-population generalization.** Generalization across different CT scanners and patient populations could be enhanced by collecting multicenter data with diverse scanning parameters and demographics, combined with domain adaptation techniques to strengthen robustness in varied clinical scenarios.
4. **Integration of clinical prior knowledge.** Incorporating clinical prior knowledge—such as patient age and symptom severity—may further refine segmentation outputs and provide more comprehensive support for clinical decision-making.
5. **Bridging specialist and generalist paradigms.** Investigating whether the proposed architectural innovations, particularly the dual-decoder anatomical context mechanism and the edge-aware loss, can be integrated into fine-tuning pipelines for foundation models such as MedSAM offers a promising direction to bridge the gap between specialist depth and generalist breadth, with significant potential for clinical translation.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code and Data Availability

All data in support of the findings of this paper are available within this article, and the code during the current study is available from the corresponding author on reasonable request.

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